SEIZE-GUARD: Real-time IoT Seizure Detection & Alert System

TEAM NAME: _COB WEB__________

Background & Introduction

- Epilepsy affects over **50 million people** worldwide, making it one of the most prevalent neurological disorders globally.
- Seizures are inherently **unpredictable** — they can occur without warning, at any time of day or night.
- Continuous caregiver supervision is **physically and emotionally unsustainable**, creating critical gaps in patient safety.
- Commercial medical-grade wearables cost **30K TO 80K**, placing them out of reach for most patients and families.
- **SEIZE-GUARD** is a low-cost, ESP32-based wearable that delivers real-time seizure detection and emergency alerts — democratizing access to life-saving monitoring technology.

Objectives

- Design a **wearable IoT system** that alerts caregivers instantly upon seizure onset detection.
- Integrate **Telegram Bot** for zero-friction, cross-platform emergency notifications without a dedicated app.
- Continuously monitor **3-axis accelerometer data** via the MPU6050 sensor at high sampling rates.
- Implement a **smart grace-period cancel button** to reduce false positive alarms and prevent unnecessary emergency dispatches.
- Achieve a total hardware cost under Rs1000 , making the system accessible in low-resource settings.

Algorithm: Seizure Detection

Goal: Reduce false positives using a timed-threshold logic with multi-stage confirmation.

The pipeline above illustrates the sequential decision logic that filters transient motion from genuine seizure events before dispatching an emergency alert.

Signal Processing

- Calculates the **mag+**
- **magnitude** of X, Y, Z acceleration vectors in real time.
- Flags candidate events when movement magnitude exceeds **1.2G threshold**.
- Requires **sustained high movement for 500ms** before triggering a localized buzzer warning.

False Positive Mitigation

- Local **buzzer warning** activates first, providing an audible alert before cloud notification.
- User has a **5-second grace period** to press a physical cancel button.
- Telegram emergency webhook fires **only if the cancel button is not pressed** within the window.

Interface: Telegram Alert System

Instant-Messaging Webhook

Replaces a traditional mobile app with a direct Telegram Bot webhook — no installation required on the caregiver's device.

Passive Background Operation

The caregiver receives alerts passively; the system operates silently in the background without active monitoring required.

OS-Bypassing Push Notifications

Real-time push notifications bypass OS background restrictions, ensuring alerts are delivered even when the device is locked.

Results & Conclusions

- Successfully built a **fully functional hardware MVP** using the ESP32 microcontroller and MPU6050 accelerometer.
- Achieved **near-instant latency** between physical seizure-simulation movement and Telegram cloud alert delivery.
- Demonstrated **highly reliable threshold detection** during tonic-clonic seizure motion simulations with minimal false triggers.
- The cancel-button mechanism effectively **reduces nuisance alerts** without compromising emergency response speed.
- Total component cost remains under Rs 1250, validating the system's accessibility proposition.

Future Work: Miniaturization to custom PCB · Integration of TensorFlow Lite for ML-based classification .· Addition of GSM module and using pulse oximeter (MAX30102) and building app for multiple users · Clinical validation trials.

References & Acknowledgements

ZYNEX Hackathon

Project developed during ZYNEX hackathon. Thanks to organizers and mentors for infrastructure support.

Open-Source Libraries

ESP32 Arduino Core · MPU6050 I2C Library · Universal Telegram Bot Library(botfather)· and various ai tools

Academic Foundations

WHO Epilepsy Fact Sheet · datasheet from kaggle.com